Construction of the Cosmic Distance Ladder Using Observational Data

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1 Introduction

Distances in astronomy cannot be measured directly for most objects. Instead, a sequence of methods, known as the cosmic distance ladder, is used to extend distance measurements from nearby stars to distant galaxies. Each rung of the ladder calibrates the next, progressively extending the distance scale.

In this work, we construct the distance ladder using observational datasets, moving from geometric methods to empirical and dynamical relations. The ladder includes parallax, main sequence fitting, Cepheid variables, Type Ia supernovae, and galaxy scaling relations such as the Tully–Fisher and Faber–Jackson relations.

2 Parallax Distances

The most fundamental distance measurement is stellar parallax, given by

$$d = \frac{1}{p} \tag{1}$$

where p is the parallax in arcseconds and d is the distance in parsecs.

Using Gaia data, distances were computed for nearby stars and used to construct an absolute magnitude scale.

3 Main Sequence Fitting

Using stars with known distances, an absolute color-magnitude diagram (CMD) was constructed for the Hyades cluster.

$$M = m - 5\log_{10}(d) + 5 \tag{2}$$

This calibrated main sequence was then used to estimate distances to other clusters such as the Pleiades by shifting the CMD.

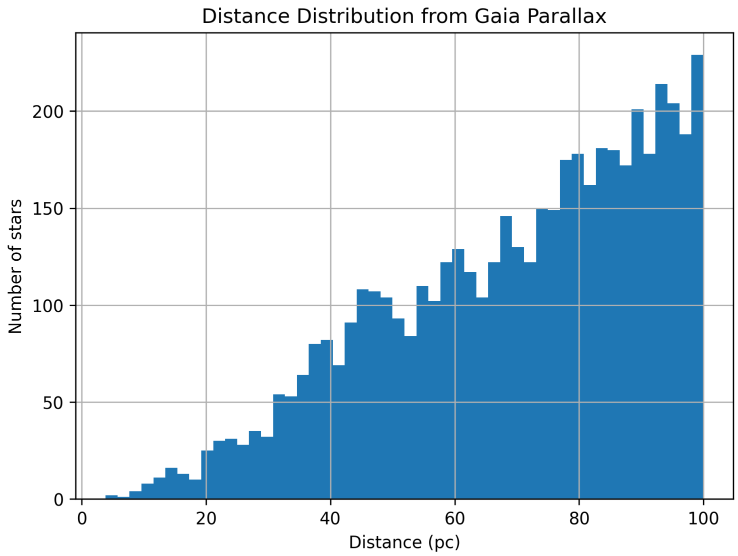

Figure 1: Distribution of stellar distances from Gaia parallaxes.

4 Cepheid Variables

Cepheid variables obey a period-luminosity relation:

$$M = a \log P + b \tag{3}$$

Light curves were analyzed to determine the period and mean magnitude.

Using the calibrated relation, the distance to the Large Magellanic Cloud (LMC) was estimated:

$$\mu = m - M - A_V \tag{4}$$

5 Type Ia Supernovae

Type Ia supernovae are standardizable candles. Their distance modulus is given by:

$$d_{\mathrm{Mpc}} = 10^{(\mu-25)/5} \tag{5}$$

Using redshift and distance:

$$v = cz \tag{6}$$

the Hubble constant was estimated.

The estimated value was:

$$H_0 \approx 64.5 \pm 8.1 \text{ km s}^{-1}\,\text{Mpc}^{-1} \tag{7}$$

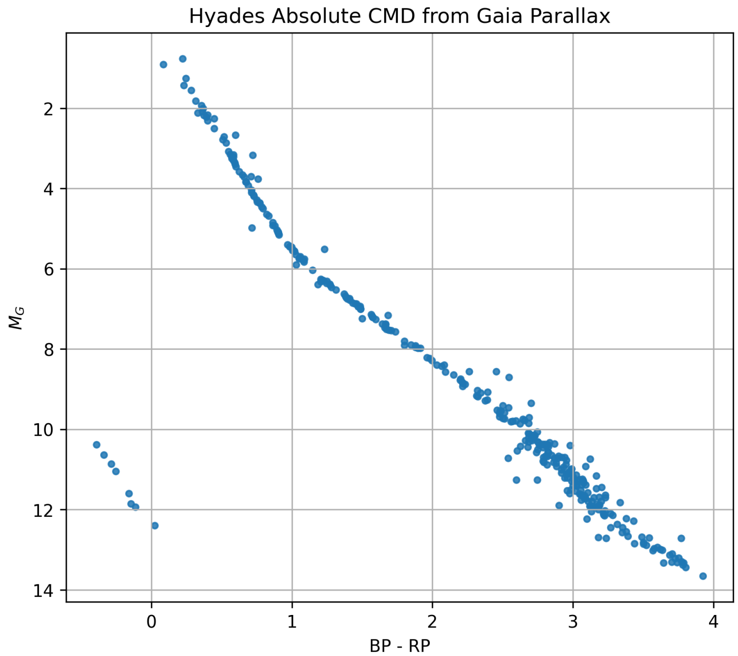

Figure 2: Absolute CMD of Hyades from parallax distances.

6 Tully–Fisher Relation

For spiral galaxies, the Tully–Fisher relation connects luminosity and rotation velocity:

$$M = a \log V + b \tag{8}$$

The rotation velocity is derived from HI 21 cm line width.
Distances computed using this relation showed strong agreement with catalog values.
Using these distances and recession velocities:

$$H_0 \approx 70.5 \text{ km s}^{-1} \text{Mpc}^{-1} \tag{9}$$

7 Faber–Jackson Relation

For elliptical galaxies:

$$L \propto \sigma^4 \tag{10}$$

or:

$$M = a \log \sigma + b \tag{11}$$

where σ is the stellar velocity dispersion obtained from spectral line broadening.
The larger scatter confirms that this method is less precise than Tully–Fisher.

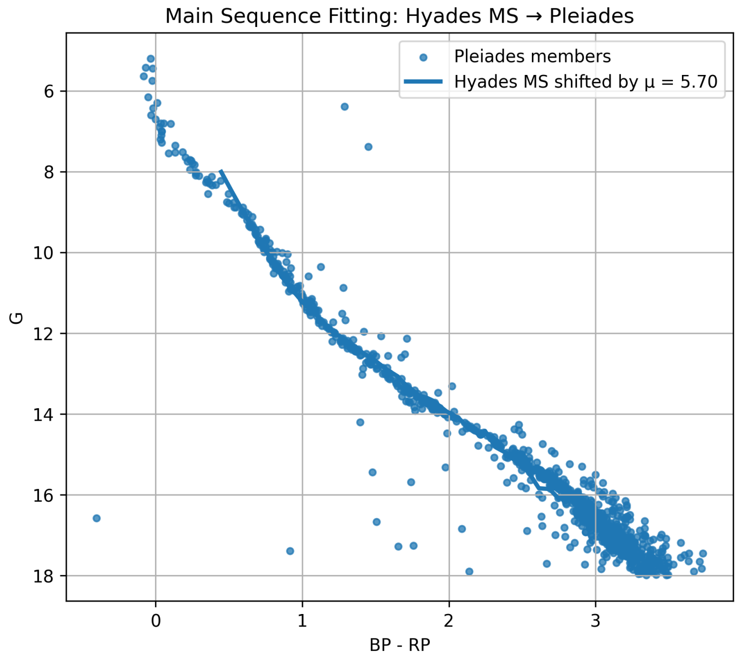

Figure 3: Main sequence fitting between Hyades and Pleiades.

8 Discussion

The cosmic distance ladder was successfully constructed using observational datasets. Each rung calibrated the next, demonstrating how absolute distance scales are built in astronomy.

Key results:

- SN Ia: $H_0 \approx 64.5$
- Tully–Fisher: $H_0 \approx 70.5$

The agreement between independent methods validates the consistency of the distance ladder.

9 Conclusion

Distances in astronomy are not measured directly but constructed through a sequence of calibrated methods. This work demonstrates the full ladder from nearby stars to cosmological scales using real data, highlighting both the power and limitations of each method.

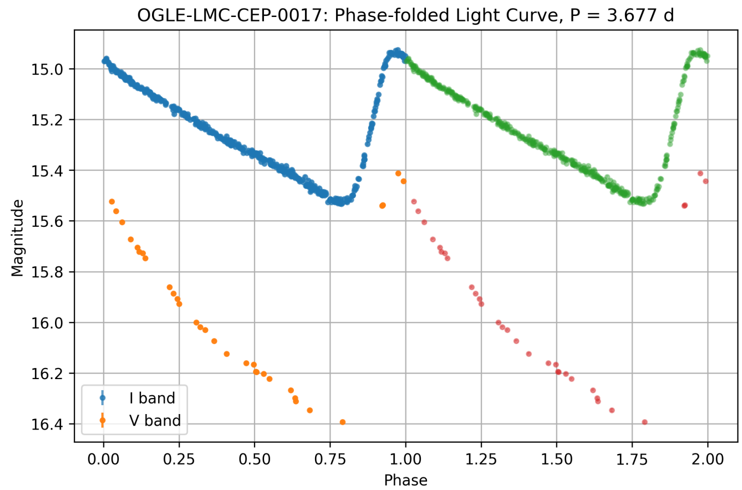

Figure 4: Phase-folded light curve of Cepheid variable.

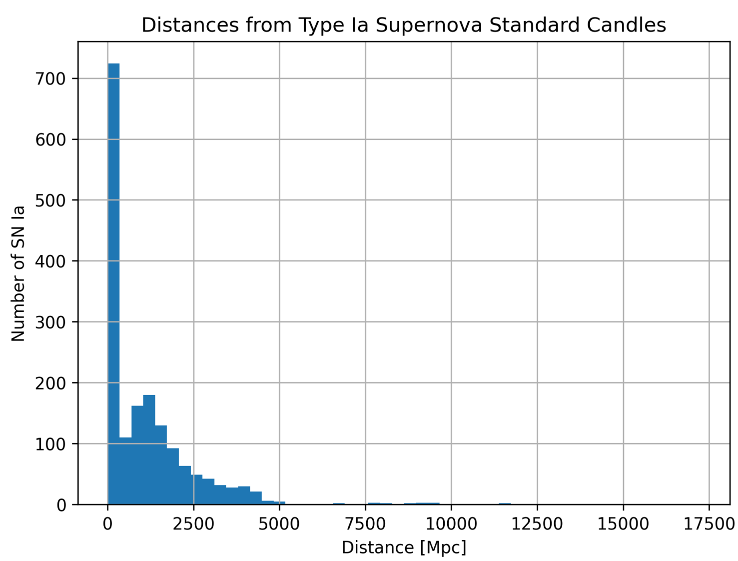

Figure 5: Distribution of SN Ia distances.

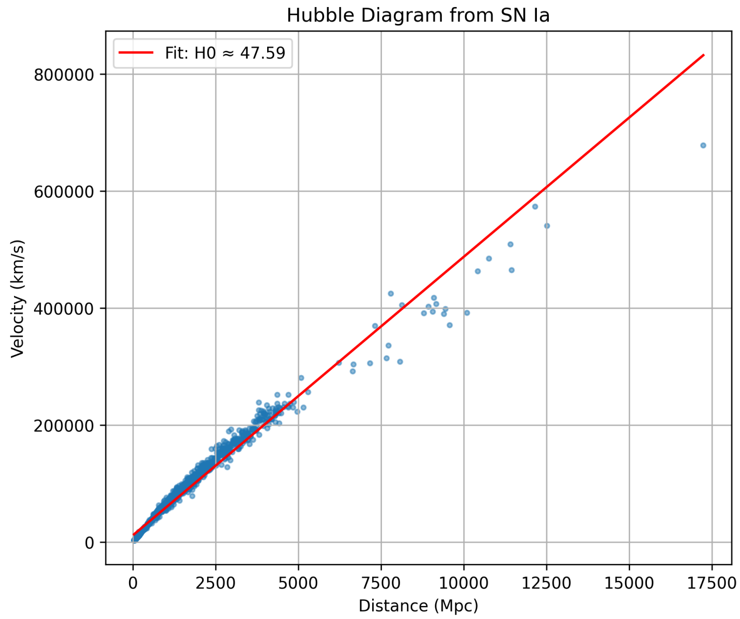

Figure 6: Hubble diagram using SN Ia distances.

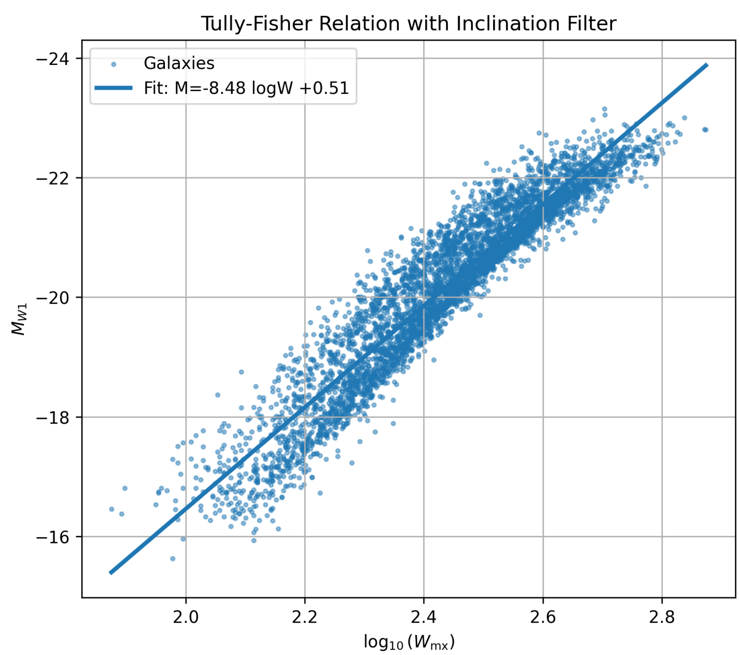

Figure 7: Tully–Fisher relation after inclination filtering.

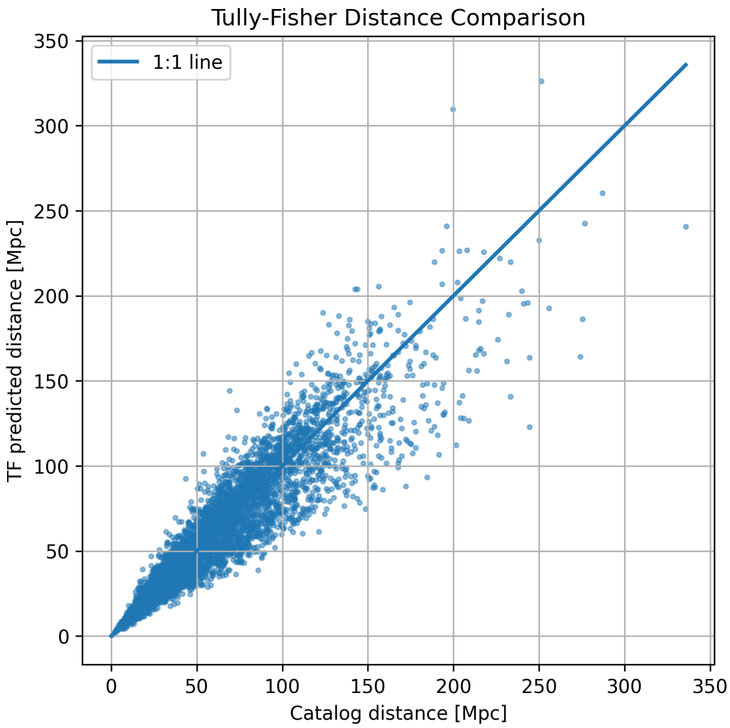

Figure 8: Comparison between TF distances and catalog distances.

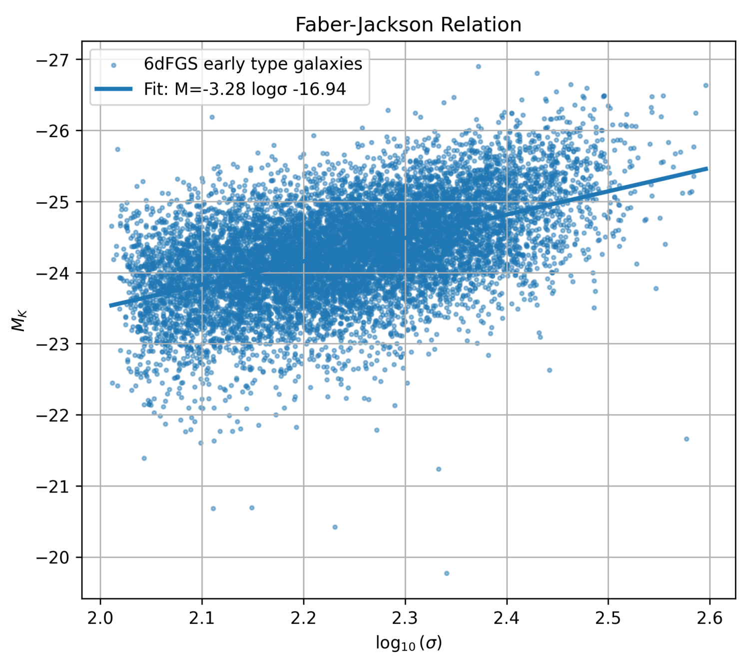

Figure 9: Faber–Jackson relation.